

MARKING SCHEME - Mathematics Quiz

Total Marks: 20 | Passing Mark: 12 | Subject: Mathematics

Note: Award marks as shown. Accept equivalent correct working methods.

Q1 [2 marks]

$$2x + 6 = 14 \Rightarrow 2x = 8 \Rightarrow x = 4$$

- 1 mark: correct rearrangement ($2x = 8$)

- 1 mark: correct final answer ($x = 4$)

Q2 [3 marks]

$$6x^2 + 9x = 3x(2x + 3)$$

- 1 mark: identifies common factor $3x$

- 2 marks: correct factorised form $3x(2x + 3)$

Q3 [4 marks]

From eq 2: $x = y + 2$ [1 mark]

Substitute: $2(y+2) + y = 10 \Rightarrow 3y = 6 \Rightarrow y = 2$ [2 marks]

Then: $x = y + 2 = 4$ [1 mark]

Final answer: $x = 4, y = 2$

Q4 [5 marks]

(a) Hypotenuse = $\sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ cm}$ [2 marks]

(b) Area = $0.5 \times 6 \times 8 = 24 \text{ cm}^2$ [1 mark]

(c) Perimeter = $6 + 8 + 10 = 24 \text{ cm}$ [2 marks]

Q5 [6 marks]

(a) $x^2 - 5x + 6 = (x - 2)(x - 3)$ [2 marks]

(b) Roots: $x = 2$ and $x = 3$ [2 marks]

(c) $x=2$: $4 - 10 + 6 = 0$ (correct) [1 mark]

$x=3$: $9 - 15 + 6 = 0$ (correct) [1 mark]